STAT 405 Project Proposal: Bayesian Hierarchical Modeling of Cancer Mutation Rates

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Team

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Repository: github.com/annmariet/stat405_project

Research Question

Somatic mutation rates for cancer-driving genes vary dramatically across cancer types. For example, the gene *TP53* is mutated in ~95% of ovarian cancers but only ~2% of kidney cancers.

Our research question is: How much do cancer risks and mutation patterns vary across cancer types, and what common structures can be learned by borrowing information across related cancers?

Main Dataset: TCGA PanCancer Atlas via cBioPortal

The Cancer Genome Atlas (TCGA) PanCancer Atlas provides uniformly processed somatic mutation data across 33 cancer types and ~11,000 tumors. We access it through the public cBioPortal API using the `cbioportalR` R package.

URL: https://www.cbioportal.org/

We will extract mutation counts for 10 well-studied oncogenes and tumor suppressors (TP53, KRAS, PIK3CA, BRAF, PTEN, APC, ARID1A, EGFR, IDH1, CDKN2A) across ~20 cancer types with $N \geq 150$ sequenced samples. Each observation is a (gene, cancer type) pair with counts (y_{gc}, n_{gc}), where y_{gc} is the number of samples harboring a mutation in gene g in cancer type c, and n_{gc} is the total number of sequenced samples.

```r
set_cbioportal_db("public")

# pull mutation data for 3 cancer types as prototype
study_ids <- c("brca_tcga_pan_can_atlas_2018",
               "luad_tcga_pan_can_atlas_2018",
               "skcm_tcga_pan_can_atlas_2018")

genes <- c("TP53", "KRAS", "PIK3CA", "BRAF", "PTEN")

pull_counts <- function(sid, genes) {
  # get sample list for this study
  samples <- available_samples(study_id = sid)
  n_total <- nrow(samples)

  # query mutations for target genes only (avoids downloading entire study)
  muts <- get_mutations_by_sample(
    sample_id = samples$sampleId,
```

```r
    study_id = sid,
    genes = genes
  )

  counts <- muts %>%
    distinct(hugoGeneSymbol, sampleId) %>%
    count(hugoGeneSymbol, name = "n_mutated")

  tibble(gene = genes) %>%
    left_join(counts, by = c("gene" = "hugoGeneSymbol")) %>%
    mutate(n_mutated = replace_na(n_mutated, 0),
           n_total = n_total,
           study = sid,
           cancer = gsub("_tcga_pan_can_atlas_2018", "", sid),
           rate = round(n_mutated / n_total, 3))
}

prototype <- bind_rows(lapply(study_ids, pull_counts, genes = genes))
```

```r
head(prototype, 15)
```

```
## # A tibble: 15 x 6
##    gene   n_mutated n_total study                        cancer  rate
##    <chr>      <int>   <int> <chr>                        <chr>  <dbl>
##  1 TP53         347    1084 brca_tcga_pan_can_atlas_2018 brca   0.32
##  2 KRAS           6    1084 brca_tcga_pan_can_atlas_2018 brca   0.006
##  3 PIK3CA       347    1084 brca_tcga_pan_can_atlas_2018 brca   0.32
##  4 BRAF           7    1084 brca_tcga_pan_can_atlas_2018 brca   0.006
##  5 PTEN          58    1084 brca_tcga_pan_can_atlas_2018 brca   0.054
##  6 TP53         295     566 luad_tcga_pan_can_atlas_2018 luad   0.521
##  7 KRAS         168     566 luad_tcga_pan_can_atlas_2018 luad   0.297
##  8 PIK3CA        28     566 luad_tcga_pan_can_atlas_2018 luad   0.049
##  9 BRAF          41     566 luad_tcga_pan_can_atlas_2018 luad   0.072
## 10 PTEN          10     566 luad_tcga_pan_can_atlas_2018 luad   0.018
## 11 TP53          72     448 skcm_tcga_pan_can_atlas_2018 skcm   0.161
## 12 KRAS          10     448 skcm_tcga_pan_can_atlas_2018 skcm   0.022
## 13 PIK3CA        22     448 skcm_tcga_pan_can_atlas_2018 skcm   0.049
## 14 BRAF         233     448 skcm_tcga_pan_can_atlas_2018 skcm   0.52
## 15 PTEN          44     448 skcm_tcga_pan_can_atlas_2018 skcm   0.098
```

```r
str(prototype)
```

```
## tibble [15 x 6] (S3: tbl_df/tbl/data.frame)
##  $ gene     : chr [1:15] "TP53" "KRAS" "PIK3CA" "BRAF" ...
##  $ n_mutated: int [1:15] 347 6 347 7 58 295 168 28 41 10 ...
##  $ n_total  : int [1:15] 1084 1084 1084 1084 1084 566 566 566 566 566 ...
##  $ study    : chr [1:15] "brca_tcga_pan_can_atlas_2018" "brca_tcga_pan_can_atlas_2018" "brca_tcga_pa
##  $ cancer   : chr [1:15] "brca" "brca" "brca" "brca" ...
##  $ rate     : num [1:15] 0.32 0.006 0.32 0.006 0.054 0.521 0.297 0.049 0.072 0.018 ...
```

The full dataset will expand this to 10 genes × ~20 cancer types ≈ 200 observations.

Backup Dataset: The Catalogue of Somatic Mutations in Cancer (COSMIC) Datasets

If the TCGA/cBioPortal approach encounters issues, we will pivot to the COSMIC (Catalogue Of Somatic Mutations In Cancer) database, a large curated resource of somatic mutation information across human cancers.

URL: https://cancer.sanger.ac.uk/cosmic/download/cosmic

```
mutant_census <- read_tsv(
  "Cosmic_MutantCensus_v103_GRCh37.tsv",
  n_max = 10
)

head(mutant_census)
```

```
## # A tibble: 6 x 26
##   GENE_SYMBOL COSMIC_GENE_ID TRANSCRIPT_ACCESSION COSMIC_SAMPLE_ID SAMPLE_NAME
##   <chr>       <chr>          <chr>                <chr>            <chr>
## 1 MYOD1       COSG18517      ENST00000250003.3    COSS2197831      TCGA-FR-A726~
## 2 MYOD1       COSG18517      ENST00000250003.3    COSS2970751      IID_H198444
## 3 MYOD1       COSG18517      ENST00000250003.3    COSS1781859      TCGA-51-4079~
## 4 MYOD1       COSG18517      ENST00000250003.3    COSS2948320      2948320
## 5 MYOD1       COSG18517      ENST00000250003.3    COSS2997007      2997007
## 6 MYOD1       COSG18517      ENST00000250003.3    COSS3001741      S0001160
## # i 21 more variables: COSMIC_PHENOTYPE_ID <chr>, GENOMIC_MUTATION_ID <chr>,
## #   LEGACY_MUTATION_ID <chr>, MUTATION_ID <dbl>, MUTATION_CDS <chr>,
## #   MUTATION_AA <chr>, MUTATION_DESCRIPTION <chr>, MUTATION_ZYGOSITY <chr>,
## #   LOH <chr>, CHROMOSOME <dbl>, GENOME_START <dbl>, GENOME_STOP <dbl>,
## #   STRAND <chr>, PUBMED_PMID <dbl>, COSMIC_STUDY_ID <chr>, HGVSP <chr>,
## #   HGVSC <chr>, HGVSG <chr>, GENOMIC_WT_ALLELE <chr>,
## #   GENOMIC_MUT_ALLELE <chr>, MUTATION_SOMATIC_STATUS <chr>
```

Methodology plan

Model 1 (naive): Independent Beta-Binomial models per gene per cancer type.

$$y_{gc} \mid \theta_{gc} \sim \text{Binomial}(n_{gc}, \theta_{gc}), \quad \theta_{gc} \sim \text{Beta}(1, 1)$$

Model 2 (hierarchical): Hierarchical Beta-Binomial with gene-level hyperparameters.

$$y_{gc} \mid \theta_{gc} \sim \text{Binomial}(n_{gc}, \theta_{gc}), \quad \theta_{gc} \sim \text{Beta}(\alpha_g, \beta_g)$$

$$\alpha_g, \beta_g \sim \text{hyperprior}$$

This enables partial pooling of mutation rates across cancer types within each gene.

Additional analyses: prior sensitivity analysis (varying hyperprior choices), posterior predictive checks, and comparison of shrinkage behavior between the naive and hierarchical models.

Team Plan

To ensure that the work is divided equally, we have created a team schedule and will meet at least once a week to make sure that we are consistently working towards deadlines.

Week	Tasks
Mar 8 - Mar 14	Determine data sets, identify research questions, and turn in proposal
Mar 15 - Mar 21	Work on literature review, develop research questions, identify gaps
Mar 22 - Mar 28	Determine naiive method and work on analysis, work on interpretation
Mar 29 - Apr 4	Determine more complex method(s) and work on analysis and interpretation
Apr 5 - Apr 11	Add discussion, conclusions, implications
Apr 12 - Apr 17	Finish report and submit